

M17 + M18 + M19 — Cost caps, Observer seat, Capacity hardening

Phase: 11–13 **Status:** DONE (2026-05-26) **Combined guide** — three smaller modules shipped in one pass for brevity. Each section below is the equivalent of an individual M-guide.

M17 — Per-encounter cost caps

Goal: configurable per-room caps on Haiku turns/minute and ElevenLabs character budget; graceful fallback on overrun.

Files:

- `portal/budgets.py` (new) — `RoomBudgetTracker`, `Decision`, `_RateWindow` sliding-60s tracker. Returns `Decision(allow, fallback, reason, remaining)` per check.
- `portal/control_room.py` — `ControlRoom.budget` property lazily builds a `RoomBudgetTracker` and keeps its caps in sync with the room's `haiku_rate_cap` / `voice_char_cap` fields.
- `portal/server.py` — `GET /api/room/budget` (usage snapshot) and `POST /api/room/budget` (set caps).

Contract on overrun:

- Haiku rate cap exceeded → `fallback='refuse'` — caller refuses the turn with a "(rate-limited)" notice.
- Voice char cap exceeded → `fallback='browser_tts'` — caller emits a 503 + `{"fallback": true}` so the browser switches to `SpeechSynthesis`.

Wire-up to v6 LLM/TTS call sites is deferred — the contract is in place; integration with `runtime.py` and `voices.py` is a follow-up so M21 LAN-test feedback can shape the latency budget. The data layer is fully testable and tested.

Tests: `tests/v7/test_voice_budget_falls_back_to_browser_tts_on_overrun.py` (5 cases), `test_haiku_rate_cap_throttles_turns_when_exceeded.py` (5 cases). 10/10 PASS.

M18 — Observer instructor seat (read-only)

Goal: a TA / preceptor login that can view every page but can't fire any mutator.

Files:

- `portal/auth.py` — adds `role` parameter to `issue_session_token(vault, role='instructor'|'observer')`, `session_role(token)` helper, `require_instructor` dependency that gates state-mutating routes.
- `portal/server.py` — `/login` now accepts a `role` form field

(defaults to 'instructor'). Twelve v7 mutating routes swapped from `auth.require_vault` to `auth.require_instructor`: `- /api/room/start`, `freeze_all`, `resume_all`, `end`, `scene_broadcast` `- /api/encounter/{id}/scene`, `assign_students` `- /api/activities` (POST), `PATCH/DELETE` `/api/activities/{id}` `- /api/room/budget` (POST) `- /api/debrief/cohort/{room_id}/notes`

Read paths (GET `/api/room/state`, GET `/api/activities`, debrief views, etc.) stay on `require_vault` — observers can read everything.

Tests: `tests/v7/test_observer_seat.py` — 5 cases (cannot freeze, sees dashboard, every v7 mutator rejected for observer, instructor default keeps mutators open, `session_role` helper default). 5/5 PASS.

Out of scope: v6's existing routes (login, character CRUD, scenarios CRUD) are NOT gated — observers can still touch the v6 library. Bigger surface area, deferred to a future pass when the observer-seat UX is validated on real LAN testing.

M19 — Capacity hardening

Goal: v1 caps — 10 concurrent encounters per room, 24 student stations. Enforce at the data + route layer; surface on the dashboard.

Files:

- `portal/control_room.py` — module constants

`MAX_ENCOUNTERS_PER_ROOM = 10`, `MAX_STUDENT_STATIONS_PER_ROOM = 24`. `CapacityExceeded` exception. `_count_student_stations` helper. `ControlRoom.add_encounter` raises on overflow.

- `portal/server.py`:

- `/api/room/start` returns 409 if `N > 10` encounters requested.

- `/portal/students/register` returns 409 if station cap would

be exceeded. `- /api/room/state` payload includes a new capacity block: `{encounters_used, encounters_max, student_stations_used, student_stations_max}` for the dashboard banner.

Tests: `tests/v7/test_capacity_caps.py` — 6 cases (data-layer 11th encounter blocked, station count helper, route 409 on big finalize, capacity block in `/state`, 25th student rejected, capacity bar updates live). 6/6 PASS.

Combined test summary

- New v7 acceptance: **21 tests across 4 files** — 10 (M17) + 5

(M18) + 6 (M19).

- **Full v7 suite: 132/132 passing** (up from 111 — +21).

- **Full v6 regression on v7: 243 passed**, same 6 env-flaky

pre-existing failures, **0 v7 regressions**.

Combined change list

Date	Module	Files
2026-05-26	M17	portal/budgets.py (new), portal/control_room.py (budget property), portal/server.py (2 routes), tests/v7/test_voice_budget_*.py, tests/v7/test_haiku_rate_cap_*.py
2026-05-26	M18	portal/auth.py (role + require_instructor), portal/server.py (login role + swap 12 routes), tests/v7/test_observer_seat.py
2026-05-26	M19	portal/control_room.py (constants + CapacityExceeded + station counter + add_encounter gate), portal/server.py (409 gates + capacity block in /state), tests/v7/test_capacity_caps.py

Known issues (combined)

- M17's wire-up to the v6 LLM call path is deferred; the data layer is testable today.

- M18 only gates v7 mutating routes. The v6 control/library surface (still on `require_vault`) is open to observers. Sweep that in a hygiene-pass module after M21 feedback.

- M19's 10 / 24 caps are v1 numbers; M21 LAN test will validate on real hardware. The constants live in `control_room` for easy tuning. A future per-room override field could let operators raise caps when they have the hardware.